Software Requirements Specification for Software Engineering: subtitle describing software

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Revision History

Date	Version	Notes
October 8	1.0	Aidan: Added sections 1.1 to 1.5 (Purpose, Scope, Stakeholders, Definitions and Acronyms, and Characteristics of Intended Reader)
October 11	1.0	Robert: Added major functional requirements
October 11	1.0	Aidan: Grouped functional requirements into categories and added functional requirements derived from the stakeholder requirements

[All capstone SRS documents should have a reflection appendix. —TPLT]

[This SRS template is based on the IEEE template, with input from co-pilot and S. Smith. It will get you started. You should not modify the section headings, without first discussing the change with the course instructor. Modification means you are not following the template, which loses some of the advantage of a template, especially standardization. —TPLT]

[Feel free to change the appearance of the report by modifying the LaTeX commands. —TPLT]

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[The SRS is not generally written, or read, sequentially. The SRS is a reference document. It is generally read in an ad hoc order, as the need arises. —TPLT]

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1 Introduction

[Explain what system is being specified. State purpose, users, and project goals. —TPLT]

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1.1 Purpose

DAIIM (Decentralized Artificial Intelligence Inference Marketplace) exists to reduce reliance on the small number of large organizations that currently provide most AI inference. It serves two groups of people.

For people who need machine learning predictions, DAIIM provides a way to obtain inferences for a task they define without trusting a single centralized provider. Each inference request is answered by multiple independent providers, and their predictions are combined into one consensus result. Because no single provider controls the outcome, the result is more reliable, harder to censor or manipulate, and not dependent on any one point of failure.

For people who build machine learning models, DAIIM provides an open marketplace where they can sell predictions from their models and earn money for accurate results. This gives independent providers a way to compete with large organizations and a reason to keep improving their models.

1.2 Scope

DAIIM allows task requesters to obtain reliable inferences by combining predictions from multiple independent inference providers.

The inputs to the system consist of task definitions, inference requests, and network participation. Task definitions are specifications of a problem to be solved via machine learning, including the expected input and output types. Inference requests ask the network to perform inference for a given task and include input data or some specification of where to find the input data. Network participation includes machine learning predictions for individual inference requests that are supplied to the network by an individual. The system outputs aggregated inference results and maintains information about the network. The aggregated inference results include the final prediction for a given inference request, as well as information about the quality of the prediction.

The expected benefits are reduced reliance on centralized AI providers, improved reliability of inferences through redundant predictions from independent providers, and incentives for providers to improve the quality, performance, availability, and cost-effectiveness of their models.

The following are outside the scope of DAIIM:

- Inference tasks whose output is not a single numeric value, such as predicting a category label. Support for additional model types is a stretch goal.
- Inference on private or sensitive data within Trusted Execution Environments (TEEs). This is a stretch goal.
- DAIIM doesn't train models. It just uses model providers' existing models to make predictions for inference requests.
- Using a custom blockchain. DAIIM will use Solana, an existing public blockchain.

1.3 Stakeholders

Primary Stakeholders

Primary stakeholders interact with DAIIM directly.

- **Task requesters**
  Task requesters are people who are requesting inferences to be made for a given task, using their own input data. These people value consensus and reliability: they want a prediction that combines the results of multiple independent models, rather than relying on a single model that may be wrong, biased, or unavailable. For example, emergency management agencies would make requests to predict river levels or wildfire risk, where a single incorrect forecast can leave a community unprepared. Similarly, decentralized finance platforms could make requests to predict stock prices, where relying on a single source of predictions creates a risk of error or manipulation. Because input data is shared with inference providers, task requesters are expected to submit data that is not sensitive.

- **Inference providers**
  Inference providers are people with computing resources and models, and want to monetize their regression models. These people usually have specialized or powerful models that can be used to make predictions for a given task.

Secondary Stakeholders

Secondary stakeholders do not use DAIIM directly, but are affected by the inferences it produces or by how it handles data.

- **Beneficiaries of machine learning inference**
  Beneficiaries are people and organizations, such as laboratories, that rely on the results of machine learning inference without requesting inferences themselves. For example, a laboratory may make decisions based on biomarker predictions that a healthcare researcher obtained through DAIIM. These people benefit from inferences that are reliable and not controlled by a single provider.

- **Regulatory bodies**
  Regulatory bodies are organizations that oversee machine learning inference and data privacy. They want to ensure that personal information and sensitive data handled by DAIIM, such as inference inputs and training data, is only accessible to the parties that need it, is encrypted, and is not retained longer than necessary to fulfill an inference request.

1.4 Definitions and Acronyms

Term	Definition
AI	Artificial Intelligence.
ML	Machine Learning.
DAIIM	Decentralized Artificial Intelligence Inference Marketplace, the system specified in this document.
Task	A problem to be solved via machine learning, such as predicting a biomarker level or a stock price.
Task definition	A specification of a task, including the expected input and output types.
Inference	A prediction produced by a machine learning model for a given input.
Inference request	A request for the network to perform inference for a given task, including input data or a specification of where to find the input data.
Task requester	A person who requests inferences to be made for a given task.
Inference provider	A person with computing resources and models who supplies predictions for inference requests to the network.
Network participation	Machine learning predictions for individual inference requests supplied to the network by an inference provider.
Aggregated inference result	The final prediction for a given inference request produced by combining the predictions of multiple inference providers, together with information about the quality of the prediction.
Ground truth	The true, observed value that a prediction is attempting to estimate.
Linear regression	A machine learning model that predicts a numeric output as a linear combination of its input features.
Decentralized	Operating without a single central organization that coordinates or controls the system.
Blockchain	A distributed, append-only ledger of transactions that is maintained by a network of nodes rather than a central authority.
On-chain	Recorded on, or executed by, a blockchain.
Web3	Applications built on blockchains, where users interact with the system through their own blockchain accounts and authorize actions by signing transactions.
Token	A digital asset on a blockchain, used by DAIIM to compensate inference providers.

TEE	Trusted Execution Environment. A secure area of a processor that protects the code and data loaded inside it from unauthorized access.

Table 1: Definitions and Acronyms

1.5 Characteristics of Intended Reader

The intended readers of this document are the software engineers, reviewers, and supervisors who will design, implement, verify, and maintain DAIIM. Readers are expected to have:

- An undergraduate-level background in software engineering, including requirements specification, software design, and testing, equivalent to the third year of a software engineering degree.
- Introductory knowledge of machine learning, including supervised learning, linear regression, model training versus inference, and basic error measures such as relative error, equivalent to an introductory undergraduate machine learning course.
- Introductory knowledge of blockchain and Web3 concepts, including blockchain transactions, accounts and wallets, transaction signing, and tokens. Familiarity with smart contract development is helpful but not required.
- Familiarity with web applications that run in a modern browser.

Terms specific to the domain of DAIIM are defined in Table 1.

2 System Overview

This section provides general information about the system. It identifies the problem, the vision, and the interfaces between the system and its environment. This section also describes user characteristics and lists the system constraints.

2.1 Problem Statement

DAIIM aims to address the limitations of relying on centralized services to provide AI inference. Centralized inference services can create single points of failure and may limit transparency into how inference results are obtained. Users may also have limited ability to independently evaluate the reliability of the results they receive. These limitations can reduce trust in AI inference services and create concerns about service availability, transparency, and the potential for biased or manipulated outputs. DAIIM seeks to provide a decentralized marketplace in which multiple inference providers contribute predictions for a requested task. The system combines these predictions to produce an aggregated result and uses provider performance to inform reputation and reward mechanisms. The desired

improvement is to provide a more transparent and resilient approach to AI inference, where users can request predictions without relying entirely on a single provider. By evaluating provider predictions and incorporating incentives for reliable participation, DAIIM aims to encourage accurate results and support accountability among participating providers.

2.2 Product Vision

DAIIM aims to make AI inference more transparent, accessible, and resilient by enabling multiple independent providers to contribute predictions through a decentralized marketplace. The system seeks to produce reliable aggregated results, encourage providers to deliver accurate predictions through reputation and reward mechanisms, and allow users to access AI inference without depending entirely on a single provider. The system's goals are to:

- Provide a marketplace where users can submit AI inference requests and receive aggregated predictions.
- Improve the reliability of inference results by evaluating and combining predictions from multiple providers.
- Encourage accurate provider participation through performance-based reputation and reward mechanisms.
- Improve transparency by making aggregated results and relevant provider quality information available to users.
- Support a decentralized approach to AI inference that reduces dependence on a single service provider.
- Provide an accessible interface for users to interact with the marketplace.

2.3 System Context

2.3.1 System Context Diagram

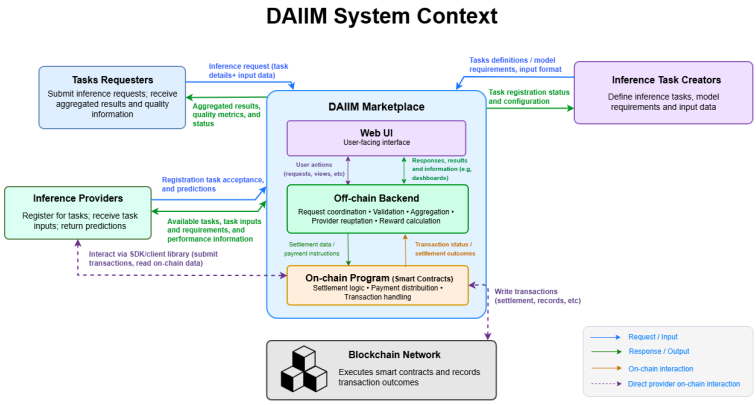


Figure 1: System Context

2.3.2 Responsibilities

- **Task Requester Responsibilities:**
  - Submit inference requests with task definitions and the required input data.
  - Provide inputs in a format supported by the system.
  - Receive aggregated inference results and relevant quality information.
- **Inference Provider Responsibilities:**
  - Receive inference tasks assigned or made available by the marketplace.
  - Execute the requested inference and return predictions in the required format.
  - Provide predictions that can be evaluated by the system's aggregation and reputation mechanisms.
- **Blockchain Network Responsibilities:**
  - Process supported on-chain marketplace transactions.
  - Maintain records of transactions and other information assigned to the blockchain.
  - Provide transaction outcomes to the system when required.
- **Software Engineering Responsibilities:**
  - Validate inference requests and detect invalid or unsupported input data.
  - Coordinate inference requests with participating providers.

- Aggregate provider predictions into a result for the requester.
- Evaluate provider performance and maintain relevant reputation information.
- Apply the marketplace's reward and penalty mechanisms according to provider performance and system rules.
- Present aggregated results and relevant quality information to users.
- Handle failed or incomplete requests without presenting them as successful results.

2.3.3 Operating Context and Safety Significance

DAIIM is intended to be used as a web-based marketplace for requesting and evaluating AI inference results. Users interact with the system through a web interface, while inference providers perform the requested computations and return predictions for aggregation. The system may interact with a blockchain network for supported marketplace transactions and records. The initial proof of concept focuses on regression-based inference and the evaluation of aggregation accuracy, provider reputation, and reward mechanisms. DAIIM is intended to support general AI inference use cases rather than directly control physical systems or make safety-critical decisions. Incorrect predictions, provider collusion, failed transactions, and inaccurate reputation or reward calculations may nevertheless reduce the reliability and trustworthiness of the marketplace. These risks should be considered when evaluating the system.

2.4 User Characteristics

DAIIM is intended to support users with varying levels of technical expertise.

- **Task Requesters:** Task requesters should have a basic understanding of the inference task they wish to perform and be able to provide the required input data. They should be able to use a modern web browser and interpret the inference results presented by the system. They are not expected to understand the internal aggregation, reputation, or reward algorithms.
- **Inference Providers:** Inference providers should have sufficient technical knowledge and computational resources to execute the inference tasks they accept. They should understand the requirements for submitting predictions in a format supported by the marketplace.
- **System Administrators or Developers:** Individuals responsible for deploying, configuring, or maintaining the system should have sufficient technical knowledge of its software components and any associated blockchain infrastructure.

The user interface should minimize the technical knowledge required to submit requests and interpret results, while provider participation may require greater technical expertise.

2.5 System Constraints

- **Web-based access:** The user-facing application must be accessible through a modern web browser and require an internet connection to communicate with the marketplace.
- **Inference execution:** The system must support the inference model or models selected for the project. The initial proof of concept focuses on regression-based inference.
- **Blockchain integration:** Any on-chain marketplace operations must be compatible with the blockchain network selected for the project. The system must account for the possibility of transaction delays or failures.
- **Input and output compatibility:** Inference requests and provider predictions must use supported data formats so that the system can validate inputs and process predictions consistently.

3 Stakeholder Requirements

This section lists the stakeholder requirements for DAIIM, grouped by the stakeholders described in Section 1.3. These requirements are technology-independent and are stated from the point of view of the stakeholders.

Task Requesters

SR1 Task requesters shall be able to define their own machine learning task, including the type of input the task takes and the type of output it produces.

SR2 Task requesters shall be able to submit an inference request for a defined task and receive a result for that request.

SR3 Task requesters shall receive inference results that are reliable. Each result shall be based on predictions from multiple independent inference providers, so that no single faulty or dishonest provider can determine the result.

SR4 Task requesters shall receive results for their inference requests even when some inference providers are unavailable.

SR5 Task requesters shall receive information about the quality of each inference result, so that they can judge how much to trust it.

SR6 Task requesters shall be able to obtain inferences without relying on a single organization to coordinate the request, supply the prediction, or control the result.

SR7 Task requesters shall be able to see the actions and payments they authorize, and verify that those actions and payments were carried out.

SR8 Task requesters shall be able to define tasks, submit inference requests, and receive results without knowledge of decentralized systems or their underlying infrastructure.

Inference Providers

SR9 Inference providers shall be able to offer predictions from their own models for tasks defined by task requesters.

SR10 Inference providers shall be able to earn a profit by supplying accurate predictions to the network.

SR11 Inference providers shall be rewarded according to the accuracy of their predictions, so that supplying honest, accurate predictions is more profitable than supplying dishonest or careless ones.

SR12 Inference providers shall receive payment for their predictions without manual intervention from the task requester or any third party.

Secondary Stakeholders

SR13 Personal information and sensitive data handled by DAIIM, such as inference inputs and training data, shall only be accessible to the parties that need it to fulfill an inference request.

SR14 Personal information and sensitive data handled by DAIIM shall be protected from unauthorized access while it is transferred and while it is stored.

SR15 Personal information and sensitive data handled by DAIIM shall not be retained longer than necessary to fulfill the inference request it belongs to.

4 User Personas

4.1 Persona 1: Emergency Management Coordinator (Task Requester)

- Name: Priya Nair
- Description: Priya is 41 and coordinates emergency management for a mid-sized town on a flood-prone river, in a region that also has an annual wildfire season. She decides when to issue flood warnings and evacuation orders, and when to pre-position firefighting resources. She currently relies on a single forecasting service, and has to justify every decision to the town council.

- Goals: Receive river-level forecasts 24 to 48 hours ahead, and wildfire risk estimates during fire season, that combine predictions from multiple independent sources. See how much the sources agree, so she knows how much to trust each forecast. Keep costs predictable and within the town's budget.
- Frustrations: Her current service is a black box: it gives one number with no indication of confidence, and she has no alternative when it is wrong or unavailable. A missed flood forecast the previous spring left the town unprepared. She has no interest in learning about cryptocurrency or managing digital wallets.
- Technical Ability: Comfortable with spreadsheets, web dashboards, and mapping tools. No programming experience and no experience with blockchain or Web3 applications.

4.2 Persona 2: Independent Stock Forecaster (Inference Provider)

- Name: Daniel Okafor
- Description: Daniel is 29 and works as a software developer. In his spare time he builds machine learning models that predict next-day stock prices from historical market data. His models perform well in backtests, but he has no way to earn money from them other than trading with his own savings.
- Goals: Earn a steady side income by supplying predictions from his models. Get paid automatically and promptly for good predictions. Build a visible track record that shows his models are reliable. Keep the cost of submitting each prediction low enough that participating is worthwhile.
- Frustrations: Selling predictions usually requires connections at a fund. Trading his own money exposes him to large losses. Prediction platforms he has tried were unclear about how payouts were calculated.
- Technical Ability: Strong Python programmer who regularly uses machine learning libraries. Owns a cryptocurrency wallet and has used decentralized finance applications.

4.3 Persona 3: Hydrology Researcher (Inference Provider)

- Name: Dr. Elena Varga
- Description: Elena is 34 and is a postdoctoral researcher in hydrology. She specializes in forecasting streamflow in mountain rivers fed by melting snow, where standard forecasting models often perform poorly during the spring melt. Her models currently only appear in academic papers.
- Goals: Earn supplementary income from her expertise by supplying predictions for tasks in her field, such as river-level and streamflow forecasts. Build a reputation for her models outside academia. See her research help communities prepare for floods.

- Frustrations: Consulting contracts are slow to arrange and rarely available to early-career researchers. Her models are most valuable during unusual events, when they disagree with standard models, and she worries about being penalized for disagreeing with the majority even when she is right.
- Technical Ability: Expert in statistical modeling, with strong Python and R skills. Has never used a cryptocurrency wallet or a blockchain application.

5 Use Cases

[Use cases are a way to describe the interactions between users and the system. They are typically written in natural language and describe the steps that a user takes to achieve a specific goal. Use cases can be used to identify functional requirements and to create test cases. —TPLT] [Typically 8-15 substantial use cases. —TPLT]

5.1 UC1: Example Use Case

Primary Actor

TODO

Preconditions

- TODO

Main Success Scenario

1. TODO
2. TODO
3. TODO

Alternative Flows

- TODO

Postconditions

- TODO

6 Functional Requirements

[Functional requirements describe what the system should do. They are typically expressed as a set of conditions or capabilities that the system must have. These requirements are usually derived from the stakeholder requirements and use cases. —TPLT]

[Use SHALL statements. Requirements must be atomic, testable, and traceable. —TPLT]

The functional requirements are grouped by the area of DAIIM they describe.

6.1 Accounts and Authorization

FR1: The system shall allow users to authenticate by connecting their Web3 wallet and signing a verification message to prove ownership of the associated wallet address.

FR2: The system shall allow unauthenticated users to browse publicly available marketplace information, including registered providers, available models, and provider reputations.

FR3: The system shall require authenticated users to authorize (sign) blockchain transactions through their connected wallet before executing actions that transfer tokens or modify on-chain state.

6.2 Inference Tasks

FR4: The system shall allow authenticated users to create new inference tasks by providing the task definition.

FR5: Each inference task definition shall specify a name, problem description, inference category or type, input schema, and output schema. Where applicable, it shall also specify the metric(s) used to evaluate predictions. The task creator may optionally provide training data, or a reference to training data, for providers to use when training or adapting models for the task.

FR6: Each inference task definition shall specify the minimum number of participating providers required to fulfill an inference request for the task.

FR7: The system shall reject inference task definitions that specify a minimum number of participating providers less than 2.

FR8: Each inference task definition shall specify a reputation tolerance T, in the units of the task's output, used to determine when a provider's reputation decreases.

FR9: Each inference task definition shall specify the length of the acceptance period for the task's inference requests. The acceptance period for each request shall end this length of time after the request is broadcast.

FR10: Each inference task definition shall specify the maximum prediction time for the task's inference requests, measured from the end of the acceptance period.

FR11: The system shall allow all users to browse existing inference tasks and view each task's definition, required minimum provider stake, and number of registered providers.

6.3 Provider Registration and Staking

FR12: The system shall allow authenticated users to register as inference providers for an inference task by staking the required minimum amount of tokens (reference the min # tokens).

FR13: When a provider registers for an inference task, the system shall lock the required minimum stake for that task in escrow. The system shall return the provider's stake when the applicable conditions for releasing it are met.

FR14: The system shall allow registered inference providers to unregister from an inference task, provided they have not accepted any acitve inference requests for that task. Upon unregistration, the provider shall be removed from the list of registered providers for the task, and their remaining staked tokens shall be returned to their token account. Upon re-registering for an inference task, the provider shall start with the baseline reputation score for that task, regardless of their previous reputation score.

FR15: The system shall allow inference providers to view their reputation score, staked tokens, and rewards earned for each inference task they are registered for.

6.4 Inference Requests

FR16: The system shall allow authenticated users to submit inference requests for an existing inference task. Each request shall include input data that conforms to the task's input schema.

FR17: The system shall reject inference requests whose input data does not conform to the task's input schema.

FR18: When submitting an inference request, the system shall require the requester to choose a payment amount in tokens greater than the minimum and prompt them to authorize, through their connected wallet, a transaction transferring that amount of tokens into escrow.

FR19: After an inference request is submitted, the system shall broadcast its task definition, proposed payment amount, acceptance-period expiry time, and maximum prediction time to all providers registered for the task.

6.5 Provider Acceptance and Predictions

FR20: Upon broadcast of an inference request, the system shall automatically begin the acceptance period, during which providers registered for the task may accept or decline participation.

FR21: At the specified expiry time, the system shall end the acceptance period. All registered providers that have not accepted the request will be treated as non-participating. Inference request acceptances from inference providers outside of the acceptance period will not be considered.

FR22: If fewer than the task's minimum number of participating providers have accepted an inference request when the acceptance period ends, the system shall cancel the request and return the requester's escrowed payment.

FR23: The system shall make the request's input data, or information needed to access it, available to all providers that accepted the request during the acceptance period.

FR24: Upon expiration of the acceptance period, the system shall begin collecting inferences from the set of providers that accepted the request. Predictions returned from inference providers that are not participating will be rejected.

FR25: The system shall allow each provider that accepted an inference request to submit a prediction for that request during the prediction phase.

FR26: The system shall accept at most one prediction from each provider for a given inference request.

FR27: The system shall reject predictions that do not conform to the task's output schema.

FR28: The system shall end the prediction phase when it has received a prediction from every provider that accepted the request or when the maximum prediction time has elapsed, whichever occurs first. The system shall not accept predictions after the phase has ended.

FR29: If fewer than the task's minimum number of participating providers have submitted predictions when the prediction phase ends, the system shall cancel the request and return the requester's escrowed payment.

6.6 Aggregation and Results

FR30: Once the prediction phase ends, the system shall aggregate the predictions into a final numerical result using a weighted aggregation algorithm in which each provider's reputation determines the weight assigned to its prediction.

FR31: The system shall return the final aggregated result, time elapsed since submitting the request and the number of inference providers that submitted predictions, to the user that submitted the inference request.

FR32: The system shall return, with each final aggregated result, the minimum and maximum of the predictions submitted for the request.

FR33: The system shall allow requesters to view the status of each inference request they have submitted and, once available, its final aggregated result.

6.7 Reputation and Rewards

FR34: After the final aggregated result for an inference request is computed, the system shall update the reputation score of each provider that submitted a prediction for the request. For each such provider i, the system shall compute d_i, the absolute difference between the provider's prediction and the final aggregated result, and $\bar{d}$, the average of d_j over all providers that submitted predictions for the request.

FR35: The system shall increase the reputation score of a provider whose distance d_i is less than or equal to the average distance $\bar{d}$.

FR36: The system shall leave unchanged the reputation score of a provider whose distance d_i is greater than $\bar{d}$ but less than or equal to $\bar{d} + T$, where T is the task's reputation tolerance.

FR37: The system shall decrease the reputation score of a provider whose distance d_i is greater than $\bar{d} + T$, where T is the task's reputation tolerance.

FR38: The system shall limit the reputation decrease a provider receives for a single inference request to at most MAX_REPUTATION_PENALTY, regardless of how far the provider's prediction is from the final aggregated result.

FR39: The system shall distribute the inference request's reward among all providers whose predictions contributed to the final aggregated result and whose reputation did not decrease as a result of the request, in proportion to each provider's reputation score. Each eligible provider i shall receive $R \cdot r_i / \sum_j r_j$, where R is the reward for the request, r_i is the reputation score used to weight provider i's prediction in the final aggregated result, and the sum is over all eligible providers.

6.8 Penalties

FR40: The system shall slash a provider's stake only under the conditions specified in FR41 and FR42, and shall burn a provider's remaining stake only under the conditions specified in FR43 and FR44.

FR41: When the system rejects a provider's prediction under FR27, the system shall slash SLASH_FRACTION of the provider's stake for the task and decrease the provider's reputation score.

FR42: When a provider that accepted an inference request has not submitted any prediction for the request by the end of the prediction phase, the system shall slash SLASH_FRACTION of the provider's stake for the task and decrease the provider's reputation score.

FR43: When a provider has been slashed for at least REPEAT_OFFENSE_COUNT of the last REPEAT_OFFENSE_WINDOW inference requests they accepted for a task, the system shall unregister the provider from the task and burn their remaining stake for the task.

FR44: The system shall automatically unregister any inference provider whose reputation score falls below the minimum reputation threshold. Upon removal, the provider's remaining staked tokens for that inference task shall be burned.

6.9 Transparency

FR45: Before prompting a user to authorize a transaction, the system shall display the action being performed and the amount of tokens being transferred.

FR46: The system shall display the status of each transaction a user has authorized until the transaction is confirmed or fails.

FR47: The system shall provide, for each transaction a user has authorized, the transaction identifier needed to look up the transaction on the blockchain.

FR48: The system shall record request payments, provider stakes, aggregated results, reputation updates, and reward distributions on the blockchain.

6.10 Data Handling

FR49: The system shall make a request's input data available only to the requester and the providers that accepted the request.

FR50: The system shall not store a request's input data on the blockchain in readable form.

FR51: The system shall delete a request's input data from any storage the system controls once the request has been completed or cancelled.

Questions / Missing reqs:
- We said that the task requester gets to choose the amount of tokens they want to offer for a request - How can we express this to inference providers in a useful way? Since the total amount does not reflect how much they will earn.
- Unless the requester also gets to choose exactly how many participants are necessary to complete the request.
- What happens if aggregation fails? Should add req for this
- What happens if not enough providers accept? Should add req for this
- Rewards distribution and provider reputation update may be changed after POC
- Need to define some of the thresholds used in reqs above: min reputation, amount to stake,

min inference request payment etc.

[The requirements are given labels so that they can be referenced in other parts of the SRS. The label names should be meaningful mnemonics, not numbers. —TPLT]

7 Data Requirements

[The Data Requirements section describes what information the system must manage. It includes the major data entities, their attributes, and the relationships between them. It also includes a data dictionary that defines the meaning of each attribute. —TPLT]

7.1 Major Data Entities

[What information must exist for the system to accomplish its goals? —TPLT]
[For each entity, provide a short explanation. —TPLT]

- User
- Song
- Session
- Report

7.2 Data Dictionary

[For each entity, define all important attributes. —TPLT]

Attribute	Type	Description
user_id	UUID	Unique identifier
email	String	User email address

Table 2: Data Dictionary

7.3 Privacy and Data Retention

[Discuss storage, retention, deletion, privacy controls, consent, encryption, and compliance requirements. —TPLT]
TODO

8 Non-Functional Requirements

[You may have to add other NFRs. —TPLT]

[List your nonfunctional requirements. You may consider using a fit criterion to make them verifiable. —TPLT]

[The goal is for the nonfunctional requirements to be unambiguous, abstract and verifiable. This isn't easy to show succinctly, so a good strategy may be to give a "high level" view of the requirement, but allow for the details to be covered in the Verification and Validation document. —TPLT]

[An absolute requirement on a quality of the system is rarely needed. For instance, an accuracy of 0.0101 % is likely fine, even if the requirement is for 0.01 % accuracy. Therefore, the emphasis will often be more on describing now well the quality is achieved, through experimentation, and possibly theory, rather than meeting some bar that was defined a priori. —TPLT]

[You do not need an entry for correctness in your NFRs. The purpose of the SRS is to record the requirements that need to be satisfied for correctness. Any statement of correctness would just be redundant. Rather than discuss correctness, you can characterize how far away from the correct (true) solution you are allowed to be. This is discussed under accuracy. —TPLT]

NFR1: **Accuracy** [Characterize the accuracy by giving the context/use for the software. Maybe something like, "The accuracy of the computed solutions should meet the level needed for <engineering or scientific application>. The level of accuracy achieved by Software Engineering shall be described following the procedure given in Section X of the Verification and Validation Plan." If you do this provide a link to the VnV plan. —TPLT]

NFR2: **Usability** Users with no knowledge of decentralized systems or their underlying infrastructure shall be able to define tasks, submit inference requests, and receive results without assistance. In usability testing with participants who have no prior blockchain or Web3 experience, at least USABILITY_COMPLETION_RATE of participants shall complete each of these tasks without assistance.

NFR3: **Security** Personal information and sensitive data handled by DAIIM, such as inference inputs, shall be encrypted while it is transferred and while it is stored by the system, so that it is protected from unauthorized access.

NFR4: **Maintainability** [The effort required to make any of the likely changes listed for Software Engineering should be less than FRACTION of the original development time. FRACTION is then a symbolic constant that can be defined at the end of the report. —TPLT]

NFR5: **Portability** [This NFR is easier to write than the others. The systems that Software Engineering should run on should be listed here. When possible the specific versions of the potential operating environments should be given. To make the NFR verifiable

a statement could be made that the tests from a given section of the VnV plan can be successfully run on all of the possible operating environments. —TPLT]

- Other NFRs that might be discussed include verifiability, understandability, reusability, performance, reliability and availability, security, and accessibility. You may have to add other NFRs.

9 AI and Machine Learning Requirements

[Refer to Dr. Mosser's guest lecture on AI requirements. —TPLT]
[Include only when applicable. —TPLT]

9.1 Model Inputs

TODO

9.2 Model Outputs

TODO

9.3 Evaluation Metrics

[Accuracy, Precision, Recall, F1, BLEU, WER, etc. —TPLT]
TODO

9.4 Known Limitations

[Explicitly document expected failure modes. —TPLT]
TODO

10 Requirements Traceability Matrices

[The Requirements Traceability Matrix (RTM) is a table that shows the relationship between different types of requirements. It is used to ensure that all requirements are traceable and that no requirements are missed. The typical flow is: Stakeholder Need -¿ Use Case -¿ Requirement —TPLT]

[There is also a traceability matrix between functional and nonfunctional requirements. The purpose of some FRs is to satisfy NFRs. Therefore, it is important to show the relationship between the two. —TPLT]

[Further traceability to test cases and modules will be given in other documents. —TPLT]

Stakeholder Req	Use Case	Functional Req
SR??	UC1	FR1

Table 3: Traceability Between Stakeholder Requirements, Use Cases, and Functional Requirements

Functional Req	Nonfunctional Req
FR??	NFR1

Table 4: Traceability Between Functional and Nonfunctional Requirements

11 Rationale

[Provide a rationale for the decisions made in the documentation. Rationale should be provided for scope decisions, functional requirements, nonfunctioanl requirements, data requirements, and AI requirements. You don't need a rationale for everything, ust the major decisions that are not obvious. You should have traceability to the entities in the document that you are referencing. —TPLT]

11.1 Provider Staking

Inference providers must stake a minimum amount of tokens to register for an inference task (FR12). The stake is locked in escrow by the system's on-chain logic, rather than held by any person, and can only be returned, slashed, or burned according to the rules in FR13, FR40, FR14, and FR44. Requiring a stake serves three purposes:

- **Resistance to fake accounts.** The final result is a reputation-weighted aggregate of provider predictions (FR30), so anyone who controls enough providers for a task can control its results. Without a stake, an attacker could register many providers at no cost, and a provider with a low reputation could abandon their account and register a new one. Requiring a stake for every registration makes each additional provider cost real tokens, which makes gaining control of a task's results expensive.

- **Penalties for misbehaviour.** A loss of reputation alone may not deter a provider who has nothing else at risk. A stake allows the system to impose a financial penalty by slashing it (FR40), or burning it when a provider's reputation falls below the minimum threshold (FR44).

- **Commitment to accepted requests.** If accepting an inference request had no cost, providers could accept every request and respond only when convenient, causing

requests to be cancelled for having too few predictions (FR29). Having a stake at risk gives providers a reason to submit a prediction for every request they accept.

A provider's stake does not determine the weight of their predictions; reputation does (FR30). This prevents providers from gaining influence over results through wealth alone, so influence must be earned through a history of predictions that agree with the network.

The minimum stake must balance two concerns. It must be high enough that registering many fake providers is expensive, but low enough that honest new providers, who have not yet earned any rewards, are not discouraged from joining.

11.2 Reputation Tolerance

Each task specifies a reputation tolerance T (FR8), and a provider's reputation only decreases when their distance from the final aggregated result exceeds the average distance by more than T (FR37). Without this margin, roughly half of the providers would lose reputation on every request, even when all predictions are close together. T is set per task because an acceptable margin depends on the units and scale of the task's output.

12 Likely Changes

LC1: The set of supported AI inference models may expand beyond the regression model used in the initial proof of concept. The inference processing and aggregation components may need to be adapted to support additional model types.

LC2: The number and behaviour of participating inference providers may change over time as providers join or leave the marketplace. The system's aggregation, reputation, and reward mechanisms may need to accommodate different levels of provider participation.

LC3: The prediction aggregation, reputation, and reward mechanisms may be refined as the proof of concept is evaluated. These changes may be required to improve prediction reliability and address issues such as inaccurate predictions, malicious providers, or collusion.

LC4: User requirements for the marketplace interface and the presentation of inference results may evolve as the system is tested. The interface may need to be updated to improve usability and make prediction quality information easier to understand.

13 Unlikely Changes

LC5: The fundamental purpose of DAIIM is unlikely to change. The system will continue to provide a marketplace for requesting AI inference and obtaining aggregated predictions from participating providers.

LC6: The system is unlikely to move away from its intended web-based form during the scope of this project. Users will continue to access the marketplace through a modern web browser with an internet connection.

LC7: The project is unlikely to introduce a physical hardware component. DAIIM will remain a software-based system that coordinates inference requests, provider predictions, and marketplace operations.

14 Appendices

14.1 Appendix A: Additional Materials

TODO

14.2 Appendix B: Development Plan

[This section is optional. It is used to explain the plan for developing the software. In particular, this section gives a list of the order in which the requirements will be implemented. In the context of a course this is where you can indicate which requirements will be implemented as part of the course, and which will be "faked" as future work. This section can be organized as a prioritized list of requirements, or it could show the requirements that will be implemented for "phase 1", "phase 2", etc. —TPLT]

14.3 Appendix C: Values of Auxiliary Constants

[Show the values of the symbolic parameters introduced in the report. —TPLT]

[The definition of the requirements will likely call for SYMBOLIC_CONSTANTS. Their values are defined in this section for easy maintenance. —TPLT]

[The value of FRACTION, for the Maintainability NFR would be given here. —TPLT]

- USABILITY_COMPLETION_RATE: the minimum fraction of usability test participants who must complete each task without assistance (NFR2). To be determined.
- MAX_REPUTATION_PENALTY: the maximum reputation decrease a provider can receive for a single inference request (FR38). To be determined.
- SLASH_FRACTION: the fraction of a provider's stake slashed for an invalid prediction or for not submitting a prediction (FR41, FR42). To be determined.
- REPEAT_OFFENSE_COUNT: the number of slashing offences that causes a provider to be unregistered (FR43). To be determined.
- REPEAT_OFFENSE_WINDOW: the number of most recent accepted inference requests over which slashing offences are counted (FR43). To be determined.

14.4 Appendix D: Generative AI Disclosure

[Provide full transparency on how generative AI was used in the creation of this document. This includes any text, figures, or tables that were generated by AI. If any skills were created for AI agents to review the documentation, those should be disclosed here. —TPLT]

14.5 Appendix E: Reflection

[This isn't part of the SRS, but it is part of the course. —TPLT]

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?

4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.

5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.

6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?